



ALMA MATER STUDIORUM
UNIVERSITÀ DI BOLOGNA
CAMPUS DI FORLÌ

91258 / B0385 Natural Language Processing

Lesson 1. Introduction

Alberto Barrón-Cedeño
a.barron@unibo.it

28/09/2026

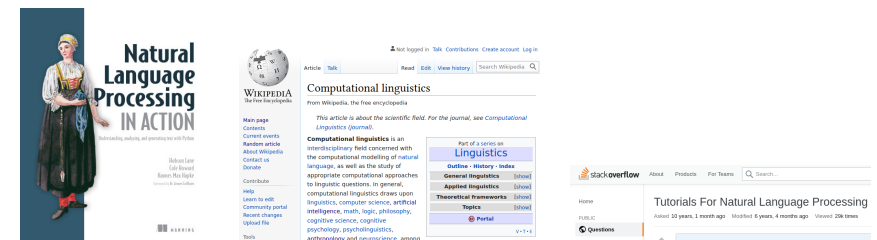
Table of Contents

1. Materials
2. Introduction
3. Requirements

Materials

Core Bibliography

1. Lane et al. (2019)'s  **Natural Language Processing in Action**¹
(a second edition is now available, including transformers)
2. Church (1994)'s **UNIX for Poets**²
(read the original)
3. Numerous **Wikipedia** articles on relevant topics
4. Multiple online forums



¹<https://www.manning.com/books/natural-language-processing-in-action>

²<https://www.cs.upc.edu/~padro/Unixforpoets.pdf>

Complementary Bibliography

1. For social media analysis
 Hovy (2021)'s *Text Analysis in Python for Social Scientists*³
2. A basic intro in Italian
 Nissim and Pannitto (2022)'s *Che cos'è la linguistica computazionale*
3. From linguistics
 Bender (2013)'s *Linguistic fundamentals for natural language processing: 100 essentials from morphology and syntax*⁴

³<https://doi.org/10.1017/9781108873352>

⁴<https://doi.org/10.2200/S00493ED1V01Y201303HLT020>

The NLP environment this year

1. Selected Topics in NLP (3 cfu)

Optional lesson covering advanced topics

 <https://www.unibo.it/it/studiare/insegnamenti-competenze-trasversali-moocs/insegnamenti/insegnamento/2026/532447>

Lesson coordinates

Slides, code, calendar and more are all available at:

 albarron.github.io/teaching/natural-language-processing

Tools

Essential

Python 3 development framework on any modern OS

1. Command line *or*
2. Integrated Development Environment; e.g., Pycharm⁵, Eclipse⁶ *or*
3. Jupyter notebook; e.g., Google's colab⁷, local Jupyter⁸

Desirable⁹

1. Git Version control system; e.g.,  Gitlab¹⁰ *or*  Github¹¹
2. $\text{\LaTeX}$ system for document preparation

⁵<https://www.jetbrains.com/pycharm/>

⁶<https://www.eclipse.org>

⁷<https://colab.research.google.com>

⁸<https://jupyter.org>

⁹Could be part of Selected topics/tutorato

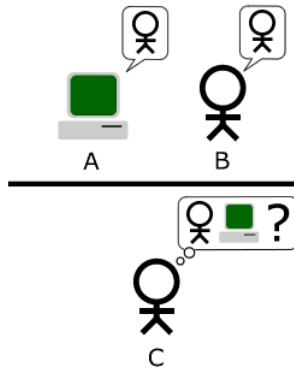
¹⁰<https://gitlab.com>

¹¹<https://github.com>

Introduction

Introduction

Natural language as a measure of intelligence



📄 Turing (1950). “Computing machinery and intelligence”. Mind. 59(236)
Picture:
upload.wikimedia.org/wikipedia/commons/e/e4/Turing_Test_version_3.png

Introduction

CL vs NLP

Computational linguistics¹²

- Interdisciplinary field concerned with the computational (it used to say “statistical or rule-based”) modeling of natural language
- Study of appropriate computational approaches to linguistic questions

Natural Language Processing¹³

- The processing of natural language information by a computer [...] Subfield of computer science, closely associated with artificial intelligence (it used to say “linguistics”)
- Data is collected in text corpora, using either rule-based, statistical or neural-based approaches in machine learning and deep learning¹⁴

¹²https://en.wikipedia.org/wiki/Computational_linguistics

¹³https://en.wikipedia.org/wiki/Natural_language_processing

¹⁴From an old version

Introduction

CL vs NLP

Natural Language Processing (Lane et al., 2019, p. 4)

- Area of research in computer science and artificial intelligence concerned with processing natural languages
- This processing generally involves translating natural language into data (numbers) that a computer can use to learn about the world

What started as an effort to translate between languages [in the 1950s] evolved into a much wider field of natural language processing.¹⁵

¹⁵https://en.wikipedia.org/wiki/Computational_linguistics

Introduction

Rule-based vs Statistical NLP

Introduction

Rule-based NLP

Models are based on a number of hand-crafted rules or grammars



Diagram borrowed from L. Moroney's Introduction to TensorFlow for Artificial Intelligence, Machine Learning, and Deep Learning

Introduction

Rule-based NLP

Models are based on a number of hand-crafted rules or grammars

```
greeting_inputs = ("hey", "morning", "evening", "hi",
                   "whatsup", "hello")
greeting_responses = ["hey", "hey hows you?", "*nods*",
                      "hello, how you doing", "hello",
                      "Welcome, I am good and you"]
```

```
def generate_greeting_response(input):
    for token in input.split():
        if token.lower() in greeting_inputs:
            return random.choice(greeting_responses)
```

<https://stackabuse.com/python-for-nlp-creating-a-rule-based-chatbot/>

Introduction

Statistical NLP

Models are *tuned* on *annotated* data

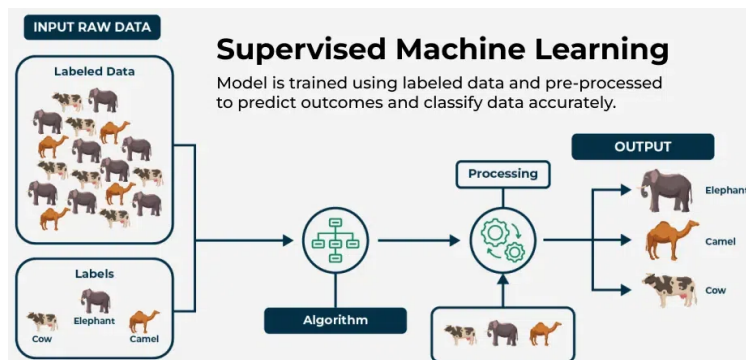
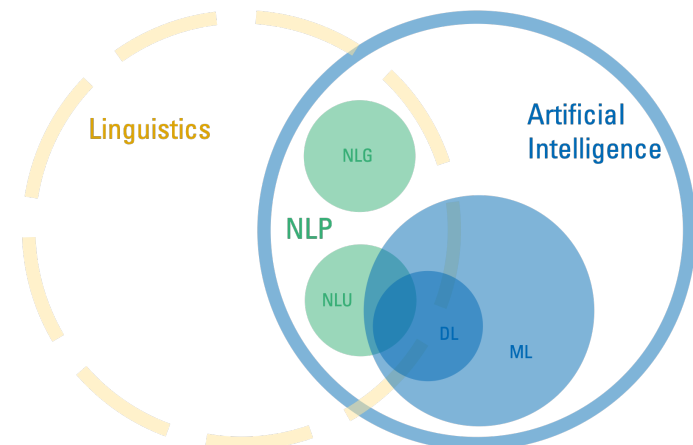


Diagram borrowed <https://www.geeksforgeeks.org/machine-learning/supervised-machine-learning>

Introduction

The NLP neighborhood



Borrowed from
<https://www.retresco.de/en/how-to-ai-natural-language-processing/> (2025)

Introduction

Non-exhaustive list of NLP applications with examples

 Search	web search engines · text autocompletion
 Editing	grammar issues identification
 Dialogue	chatbot creation
 Email	spam filtering · message classification
 Text mining	(multi-)document summarisation
 News analysis	event identification · fact checking
 Forensics	plagiarism detection · authorship attribution
 Sentiment analysis	product review ranking · opinion mining
 Creative writing	text generation with a narrative and style
 Translation	translation · quality estimation

Partially derived from (Lane et al., 2019, p. 8)

The philosophy of this lesson

1. Concept understanding
2. Know what you are doing and why
3. In position to go forward

Requirements & Evaluation

Requirements

Necessary

- Linguistics
- Algebra
- Programming in Python

Desirable

- Intermediate programming (e.g., object-oriented, testing)
- High-performance computing (e.g., slurm)¹⁶

¹⁶Mostly for seINLP

Evaluation

Final project: 80%

You will address a relevant problem. . .

- within the range of your own (research) interests **or**
- participating (formally) in a shared task **or**
- proposed by me, if you prefer

Homework: 20%

- mostly programs addressing relatively small problems

Evaluation

Typical final project pipeline

1. You propose a topic/problem. We assess if it is reasonable, doable. . .
2. You [compile data,] study the problem, design experiments, code. . .
IF you plan for a publication¹⁷
 - We meet regularly to see the advances and shape the experiments, submissions, and/or paper towards the submission deadline**ELSE**
 - We could meet sporadically, if you need it
3. You submit a written report (~ 7 pages),¹⁸ your implementation, results **1 week before the appello**
4. We meet on the date of the appello to discuss about your project, in the context of the lecture

¹⁷Talk to me well in advance; it would require my heavy involvement

¹⁸I do not like (student) novels

Evaluation

Final mark

(Beside the 20% for homework) a combination of the quality of the experiments, report, code, and oral discussion

Targeting 30L?

If I let you submit a paper, it is very likely. But it is not the only way. . .

$$p(30L \mid \text{paper submitted} == \text{True}) \approx 0.85 \quad (1)$$

$$p(30L \mid \text{paper submitted} == \text{False}) \approx 0.15 \quad (2)$$

Limited use of AI is permitted for specific tasks, such as spelling and grammar check. The student will be asked to submit a statement indicating how AI was used. Generative use of AI to actually produce work is strictly forbidden. Any violation of this rule will be considered a serious breach of academic ethics.

Evaluation

Previous final projects

2024–2025

 Benchmarking Bilingual Text Anonymization and Automatic Term Extraction Approaches

2023–2024

 Multilingual Persuasion Technique Detection in News *♠
 Tracing source languages in Wikipedia articles

2022–2023

 Sentiment analysis of video game reviews
 Authorship attribution: machine vs human

2021–2022

 Hate Speech Detection in Incel Online Spaces *♠
 Fishing for catfishes: predicting the author gender in Reddit

* student with previous programming experience

Evaluation

Previous final projects

2020–2021

- ✂ Semantic similarity between originals and machine translations •
- 🔑 Definition extraction on food-related Wikipedia articles •
- ☰ Identifying Characters' Lines in Original and Translated Plays •
- 🐦 Classifying an Imbalanced Dataset with CNN, RNN, and LSTM •

2019–2020

- ♥ AriEmozione: Identifying Emotions in Opera Verses *♣
- 🐦 UniBO@AMI: A Multi-Class Approach to Misogyny and Aggressiveness Identification on Twitter Posts Using ALBERTo *♣

* students with previous programming experience

- turned into (part of a) thesis ♣ turned into a publication

Visit the [projects section](#) of the class website for details, reports and papers

References

- Bender, E. M.
2013. *Linguistic Fundamentals for Natural Language Processing: 100 Essentials from Morphology and Syntax*. Morgan & Claypool Publishers.
- Church, K.
1994. *UNIX for poets*.
- Hovy, D.
2021. *Text Analysis in Python for Social Scientists: Discovery and Exploration*, Elements in Quantitative and Computational Methods for the Social Sciences. Cambridge University Press.
- Lane, H., C. Howard, and H. Hapkem
2019. *Natural Language Processing in Action*. Shelter Island, NY: Manning Publication Co.
- Nissim, M. and L. Pannitto
2022. *Che cos'è la linguistica computazionale*. Carocci editore.